

Lesson 4 · Anti-differentiation, Rates and Kinematics

Learning intention

Anti-differentiate and find c , interpret rates of change in context, and move between displacement, velocity and acceleration.

Success looks like

- I can find $f(x)$ from $f'(x)$ and a point.
- I can find a rate of change and say what it means, with units.
- I can go from s to v to a , and back again.

Do Now

1 The graph of f has a maximum at $x = -1$ and a minimum at $x = 4$.

Where does the graph of f' cross the x -axis?

2 $f'(x)$ is positive for $x < 2$ and negative for $x > 2$. Does f have a maximum or a minimum at $x = 2$?

3 Differentiate $s = 5t^2 - 3t + 8$.

4 Find c if $12 = 2(3)^2 - 4 + c$.

The one idea

Anti-differentiation **undoes** differentiation.

$$x^n \longrightarrow \frac{x^{n+1}}{n+1} + c$$

Raise the power by 1, divide by the **new** power, add $+c$.

Then use a known point to find c .

Check: differentiate your answer; you must get back $f'(x)$.

Worked examples · anti-differentiate

1 Find $f(x)$ if $f'(x) = 6x^2 - 4x + 5$.

2 $f'(x) = 4x - 3$ and the curve passes through $(2, 7)$. Find $f(x)$.

Step 1: anti-differentiate

Step 2: substitute the point

Step 3: c

Practice

- 1 Anti-differentiate $f'(x) = 9x^2 + 2x - 7$.
- 2 Anti-differentiate $f'(x) = x^3 - 6x$.
- 3 $f'(x) = 6x - 5$ and $f(1) = 4$. Find $f(x)$.

Rates of change

Write it in this order

1. Differentiate: the derivative is the rate.
 2. Substitute the time. (Given a *value*? Solve for the time first.)
 3. Units “per ...”; positive = increasing, negative = decreasing.
- 3** The water in a tank is $V = 2000 + 30t - 0.5t^2$ litres after t minutes. Find the rate of change at $t = 10$ and at $t = 40$, and say what each means.

Worked example · find the time

4 A population is $P = 500 + 40t - t^2$ after t years. When is it growing at 10 per year?

Set the derivative equal to the rate, then solve.

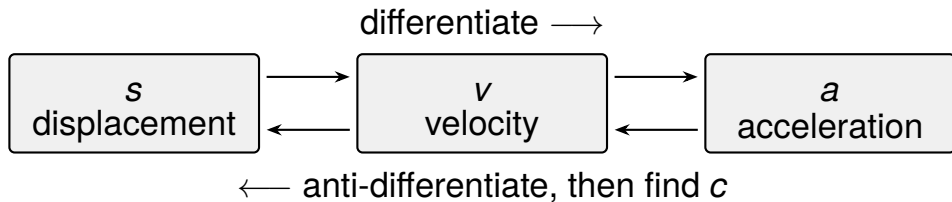
Sentence frame

At $t = \dots$ the ... is increasing / decreasing at ... per ...

Practice

- 1 The temperature of a drink is $T = 80 - 6t + 0.1t^2$ degrees Celsius after t minutes. Find the rate of change at $t = 10$ and explain what it means.
- 2 The number of people in a queue is $N = 15t - t^2 + 20$ after t minutes. When is the number in the queue not changing?
- 3 The area of a weed patch is $A = t^2 - 4t + 10$ square metres after t weeks ($t \geq 2$). Find the rate at which the area is growing when the area is 31 square metres.

Kinematics



words

maths

at rest, changes direction, maximum height

starts from the origin

starts from rest

Worked examples · motion

- 5** A particle moves with $s = t^3 - 6t^2 + 9t$ metres after t seconds. Find its velocity at $t = 2$, the times when it is at rest, and its acceleration at $t = 3$.
- 6** A ball is thrown up from 1.2 m above the ground at 14 m s^{-1} . Its acceleration is -9.8 m s^{-2} . Find its maximum height.

Practice

- 1 A particle moves with $s = 2t^3 - 9t^2 + 12t$ metres after t seconds. When is it at rest?
- 2 A cyclist starts from rest at the start line with acceleration $a = 0.6t \text{ m s}^{-2}$. Find the cyclist's velocity and distance from the start line after 5 seconds.
- 3 A stone is thrown straight up from the ground at 24.5 m s^{-1} ($a = -9.8 \text{ m s}^{-2}$). Find its maximum height and when it lands.

Ways to lose marks

- L1 $+c$, then find it from the given point.
- L2 Divide by the **new** power.
- L3 A rate is the derivative, not the value.
- L4 Say increasing or decreasing, with units.
- L5 Maximum height and at rest: $v = 0$.

Summary

1. $x^n \rightarrow \frac{x^{n+1}}{n+1} + c$; a point gives c .
2. Rate of change = derivative; say what the sign means, with units.
3. $s \rightarrow v \rightarrow a$: differentiate. $a \rightarrow v \rightarrow s$: anti-differentiate.
4. At rest or maximum height: $v = 0$.

Practice sheet · 30 minutes

Work alone on the practice sheet.

- Questions 1 and 2: warm-up, about 6 minutes.
- Questions 3 to 8: exam questions, about 24 minutes.
- Every rate and every motion answer needs units.

Homework

Homework sheet: 14 questions, about 60 minutes

- Questions 1 to 11: Achieved practice.
- Questions 12 and 13: Merit.
- Question 14: Extension (Excellence).

Optional extra: StudyTime Guide, “Quick Questions” on page 29.